

Matematica - Clasa a 7-a

Exercitii: Capitol II - Operatii cu numere reale

Nume: _____

Total: 100p

Data: _____

Sețiunea 1 - Calcule cu radicali (simplifica la maximum)

1. Produse și caturii:

$\sqrt{3} \times \sqrt{3} = ______$

$\sqrt{5} \times \sqrt{5} = ______$

$\sqrt{2} \times \sqrt{8} = ______$

$\sqrt{3} \times \sqrt{12} = ______$

$\sqrt{6} \times \sqrt{6} = ______$

$2\sqrt{3} \times 3\sqrt{3} = ______$

$4\sqrt{2} \times 2\sqrt{2} = ______$

$\sqrt{5} \times 2\sqrt{5} = ______$

$\sqrt{50} \div \sqrt{2} = ______$

$\sqrt{18} \div \sqrt{2} = ______$

$\sqrt{75} \div \sqrt{3} = ______$

$\sqrt{48} \div \sqrt{3} = ______$

2. Adunări și scăderi (scoate factorii mai întâi!):

$\sqrt{12} + \sqrt{27} = ______$

$\sqrt{50} - \sqrt{18} = ______$

$\sqrt{75} - \sqrt{48} = ______$

$2\sqrt{8} + 3\sqrt{2} = ______$

$3\sqrt{12} - \sqrt{75} = ______$

$\sqrt{32} + \sqrt{8} - \sqrt{18} = ______$

$\sqrt{45} - \sqrt{20} + \sqrt{5} = ______$

$2\sqrt{18} - 3\sqrt{8} + \sqrt{32} = ______$

Sețiunea 2 - Produse remarcabile cu radicali

3. Calculează folosind formulele $(a+b)(a-b)=a^2-b^2$ și $(a+b)^2=a^2+2ab+b^2$:

$(\sqrt{7}+1)(\sqrt{7}-1) = ______$

$(\sqrt{5}+2)(\sqrt{5}-2) = ______$

$(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2}) = ______$

$(\sqrt{2}+1)^2 = ______$

$(\sqrt{3}-1)^2 = ______$

$(\sqrt{5}+\sqrt{2})^2 = ______$

$(2+\sqrt{3})^2 = ______$

$(\sqrt{6}-\sqrt{2})(\sqrt{6}+\sqrt{2}) = ______$

$(3-\sqrt{5})(3+\sqrt{5}) = ______$

Exercitii: Capitol II (continuare)

Sectionea 3 - Rationalizarea numitorului

4. Rationalizeza (elimina radicalul din numitor):

$$1/\sqrt{2} = ______$$

$$1/\sqrt{3} = ______$$

$$2/\sqrt{5} = ______$$

$$3/(2\sqrt{3}) = ______$$

$$6/\sqrt{6} = ______$$

$$4/\sqrt{8} = ______$$

$$5/\sqrt{10} = ______$$

$$\sqrt{2}/\sqrt{6} = ______$$

5. Rationalizeza prin conjugat:

$$1/(1+\sqrt{2}) = ______$$

$$1/(1-\sqrt{3}) = ______$$

$$2/(3+\sqrt{5}) = ______$$

$$(\sqrt{3})/(2-\sqrt{3}) = ______$$

$$1/(\sqrt{5}+\sqrt{3}) = ______$$

$$(\sqrt{2}+1)/(\sqrt{2}-1) = ______$$

Sectionea 4 - Puteri cu exponent intreg

6. Calculeaza puterile:

$$2^{(-3)} = ______$$

$$3^{(-2)} = ______$$

$$5^0 = ______$$

$$(1/2)^{(-3)} = ______$$

$$4^{(-1)} = ______$$

$$(2/3)^{(-2)} = ______$$

$$(-2)^{(-3)} = ______$$

$$10^{(-2)} = ______$$

7. Aplica regulile de calcul:

$$2^3 \times 2^4 = ______$$

$$3^5 \div 3^2 = ______$$

$$(2^2)^3 = ______$$

$$5^2 \times 5^{(-2)} = ______$$

$$2^{(-1)} + 2^{(-2)} = ______$$

$$3^2 - 3^0 - 3^{(-1)} = ______$$

$$(1/2)^3 \times 2^5 = ______$$

$$4^{(1/2)} \times 4 = ______$$

Exercitii: Capitol II (continuare)

Sectiunea 5 - Ordinea operatiilor (rezolva pas cu pas)

8. Calculeaza:

a) $\sqrt{12} + 2\sqrt{3} \times \sqrt{3} - \sqrt{75} =$

b) $(\sqrt{2} + 1)^2 - \sqrt{2} \times (\sqrt{2} + 3) =$

c) $2^3 + 4 \times 2^{-1} - \sqrt{9 \times 4} =$

d) $3\sqrt{2} \times \sqrt{8} - (\sqrt{5} - 1)(\sqrt{5} + 1) =$

e) $(\sqrt{50} - \sqrt{18}) \div \sqrt{2} + (\sqrt{3} + 1)(\sqrt{3} - 1) =$

f) $2\sqrt{12} - \sqrt{3} \times (2 + \sqrt{3}) =$

Exercitii: Capitol II (continua)

Sectionea 6 - Expresii algebrice si simplificari

9. Simplifica la forma $a + b\sqrt{c}$ sau $a\sqrt{c}$:

a) $\sqrt{75} + \sqrt{48} - 2\sqrt{12} =$

b) $(2+\sqrt{3})^2 - (2-\sqrt{3})^2 =$

c) $(3+\sqrt{5})(3-\sqrt{5}) + \sqrt{5} \times \sqrt{20} =$

d) $\sqrt{6}(\sqrt{6} + \sqrt{3}) - \sqrt{18} =$

e) $1/\sqrt{2} + \sqrt{2}/2 - \sqrt{8} =$

f) $(\sqrt{3}+\sqrt{2})^2 + (\sqrt{3}-\sqrt{2})^2 =$

Sectionea 7 - Adevarat sau Fals (justifica!)

• $\sqrt{2}+\sqrt{3} = \sqrt{5}$ _ _

• $(\sqrt{a}+\sqrt{b})^2 = a+b$ _ _

• $\sqrt{a} \times \sqrt{a} = a$ ($a \geq 0$) _ _

• $1/\sqrt{2} = \sqrt{2}/2$ _ _

• $(\sqrt{3}+1)(\sqrt{3}-1) = 2$ _ _

• $2^{-2} = -4$ _ _

• $\sqrt{a^2+b^2} = a+b$ _ _

• $(a^m)^n = a^{(m+n)}$ _ _

Exercitii: Capitol II (continua)

Sectionea 8 - Probleme si calcule finale

10. Calcul rapid - scrie doar rezultatul:

$$(\sqrt{11}+1)(\sqrt{11}-1)=---$$

$$(\sqrt{7})^2 \times (\sqrt{2})^2 = ---$$

$$\sqrt{50} \div \sqrt{2} \times \sqrt{8} = ---$$

$$(1+\sqrt{2})(1-\sqrt{2})+3=---$$

$$2^3 \times 2^{-3} + 5^0 = ---$$

$$\sqrt{3}(\sqrt{27}-\sqrt{3})=---$$

11. Rezolva complet (arata toti pasii):

a) Rationalizeaza si simplifica: $(2+\sqrt{6})/(2-\sqrt{6})$

b) Simplifica: $(\sqrt{5+\sqrt{3}})^2 - (\sqrt{5-\sqrt{3}})^2$

c) Calculeaza: $\sqrt{2+\sqrt{3}} \times \sqrt{2-\sqrt{3}}$

12. Demonstreaza ca expresia este rationala:

$$E = (1+\sqrt{2+\sqrt{3}})(1+\sqrt{2-\sqrt{3}})$$

(Indicatie: grupeaza $(1+\sqrt{2})$ si aplica $(a+b)(a-b)$)

$$E = ---$$